

Find and Fix: Parameters, Polar Bounds, and Speed

Directions

- Each problem shows work that is wrong in one specific way. Repair it, then say what rule was broken.
- Part (a) always wants a corrected value; put it on the Answer line. Part (b) wants a sentence or two, written in the work space.
- Angles are in radians throughout.

1. A curve is given by $x = t^2$, $y = t^3 - 3t$. A student is asked for $\frac{dy}{dx}$ at $t = 2$ and reports 9, having computed $\frac{dy}{dt} = 3t^2 - 3$.

- (a) Find the correct value of $\frac{dy}{dx}$ at $t = 2$.
- (b) Name the error the student made.

Answer: _____

2. A particle in the plane has velocity components $x'(t) = 3t^2$ and $y'(t) = 4t$. Asked for its speed at $t = 1$, a student reports 7.

- (a) Find the correct speed at $t = 1$.
- (b) Name the error, and say why the components cannot simply be added.

Answer: _____

3. To find the area enclosed by the polar curve $r = 4\sin\theta$, a student writes $A = \frac{1}{2} \int_0^{2\pi} 16\sin^2\theta \, d\theta$ and obtains $8\pi \approx 25.133$.

- (a) Find the correct area. Give an exact answer and a decimal to three places.
- (b) Explain what went wrong with the limits, and describe how to check the correct interval.

Answer: _____

4. For the curve $x = t^2$, $y = t^4$ (with $t > 0$) a student correctly finds $\frac{dy}{dx} = 2t^2$, then differentiates that with respect to t and reports $\frac{d^2y}{dx^2} = 12$ at $t = 3$.

(a) Compute $\frac{d}{dt} \left(\frac{dy}{dx} \right)$, then find the correct value of $\frac{d^2y}{dx^2}$ at $t = 3$.

(b) Name the step the student omitted.

Answer: _____

5. Asked for the area of one petal of the polar rose $r = 3 \cos(3\theta)$, a student integrates over a full turn and reports $\frac{9\pi}{2} \approx 14.137$.

(a) Find the correct area of one petal. Give an exact answer and a decimal to three places.

(b) Explain how the correct limits are found from the equation itself.

Answer: _____

6. A particle has velocity components $x'(t) = 2t$ and $y'(t) = 3t^2$ for $0 \leq t \leq 2$. Asked for the total distance travelled, a student computes $\int_0^2 2t \, dt + \int_0^2 3t^2 \, dt = 4 + 8 = 12$.

- (a) Find the correct total distance, to three decimal places.
- (b) Name the error the student made.

Answer: _____

7. For the polar curve $r = 1 + \cos \theta$, a student is asked for the slope of the tangent line at $\theta = \frac{\pi}{2}$. They compute $\frac{dr}{d\theta} = -\sin \theta$, evaluate it at $\frac{\pi}{2}$, and report a slope of -1 .

- (a) Find the correct slope $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$.
- (b) Explain why $\frac{dr}{d\theta}$ is not the slope of the tangent line.

Answer: _____

8. A particle has position $x(t) = t^2 - 4t$, $y(t) = \frac{t^3}{3} - t$. Asked for the speed at $t = 3$, a student evaluates the position at $t = 3$, gets $(-3, 6)$, and reports $\sqrt{45} \approx 6.708$.

- (a) Find the correct speed at $t = 3$, to three decimal places.
- (b) Name the error, then decide whether the particle is speeding up or slowing down at $t = 3$ and justify your answer.

Answer: _____